

HTTP/1.1 vs HTTP/2 - Beginner Friendly Notes

What is HTTP?

HTTP (HyperText Transfer Protocol) is the protocol used for communication between a client (browser or application) and a server. It defines how requests and responses are exchanged.

HTTP/1.1

In HTTP/1.1, requests on a single connection are generally handled one after another. To improve performance, browsers often open multiple TCP connections.

HTTP/2

HTTP/2 improves performance by using a single long-lived TCP connection that supports multiple independent streams (multiplexing). It also compresses headers and uses binary framing instead of plain text.

Feature	HTTP/1.1	HTTP/2
Connection	Often multiple TCP connections	Usually one TCP connection
Requests	Sequential per connection	Multiple simultaneous streams
Protocol Format	Text	Binary
Header Compression	No	Yes (HPACK)
Multiplexing	No	Yes
Server Push	No	Yes (rarely used today)
Performance	Good	Better for many requests

Understanding Multiplexing

Imagine a highway:

HTTP/1.1: One truck enters, finishes the trip, then the next truck starts.

HTTP/2: Many trucks travel on the same highway at the same time. Each truck has its own lane identifier (stream ID), so responses can arrive in any order.

REST vs gRPC

REST is an architectural style and can use HTTP/1.1 or HTTP/2. gRPC is an RPC framework that requires HTTP/2 because it relies on multiplexing, streaming, and other HTTP/2 features.

Interview Summary

- HTTP is the communication protocol.
- HTTP/2 is a newer version with multiplexing and header compression.
- REST can run on HTTP/1.1 or HTTP/2.

- Standard gRPC uses HTTP/2.
- Multiplexing means multiple request-response streams share one TCP connection simultaneously.